

Development and testing of a data collection system for thermoregulation and thermal comfort studies

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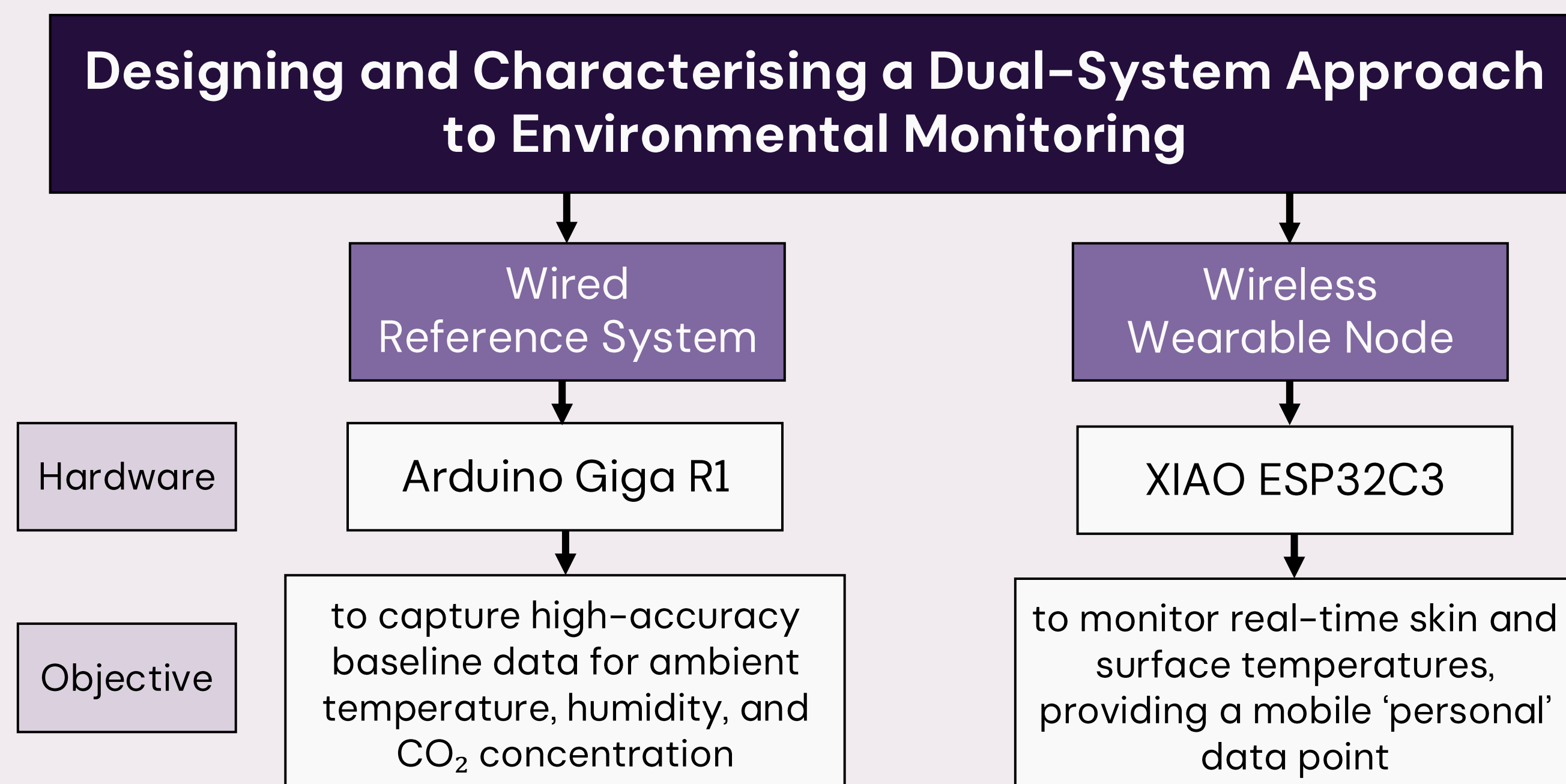
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The Problem

- Thermal comfort is a complex subjective state influenced by ambient conditions, CO₂ levels, and physiological factors.
- Traditional HVAC (Heating, Ventilation and Air Conditioning) systems rely on static wall-mounted thermostats, which often fail to account for the 'Discursive Gap' [1] between sensor readings and actual human perception.

Research Objectives



The Hypothesis

- By correlating physical sensor data (using high-precision components such as SHT45 and SDC41) with subjective user surveys across diverse UCL environments, we can define the environmental thresholds that dictate human comfort.

Experimental Design

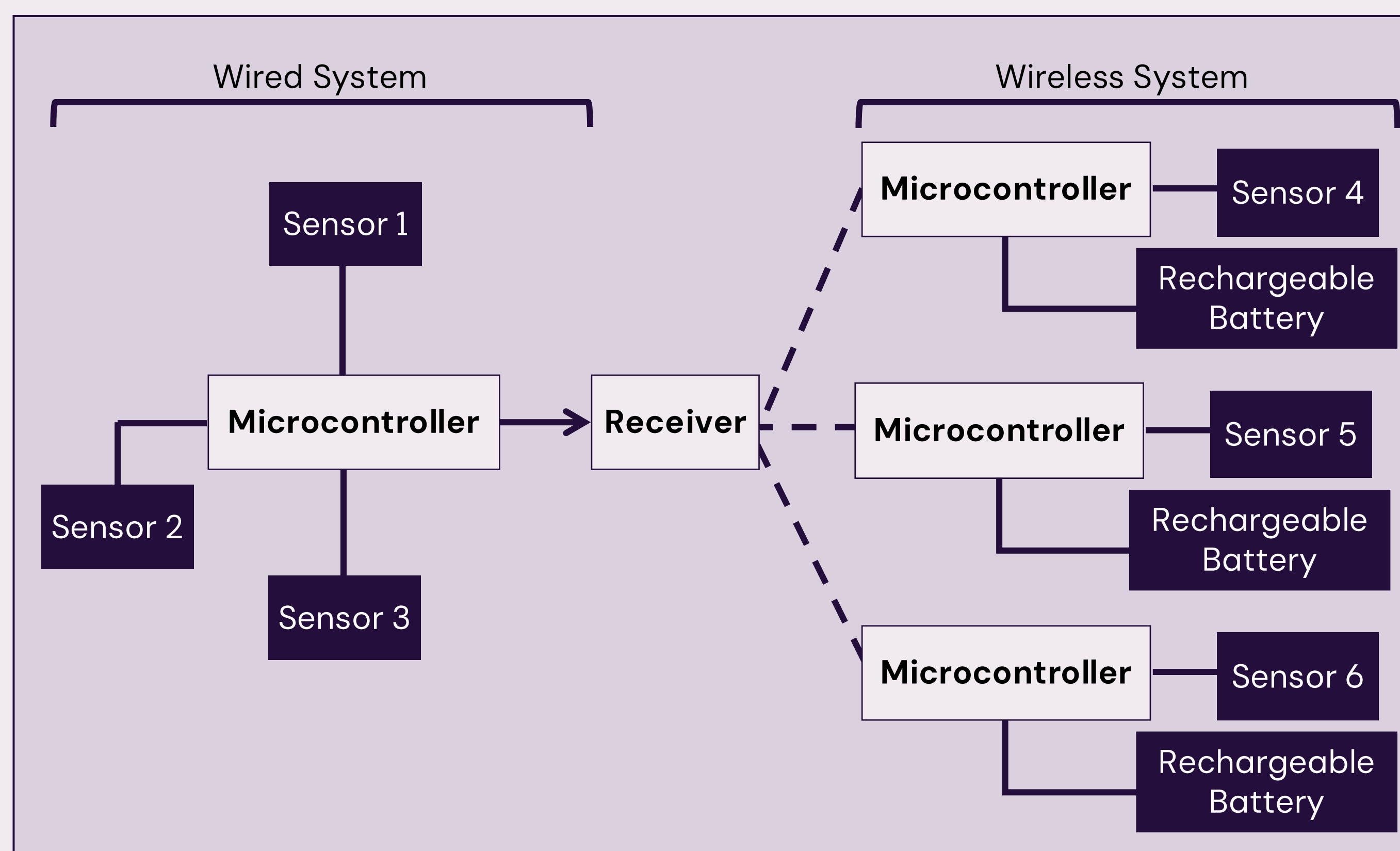


Figure 1: Conceptual design of System 2.0. It consists of wired and wireless components transmitting data to a shared receiver. All sensors are connected to the microcontrollers via Inter-Integrated Circuit (I2C) communication for its simplicity.

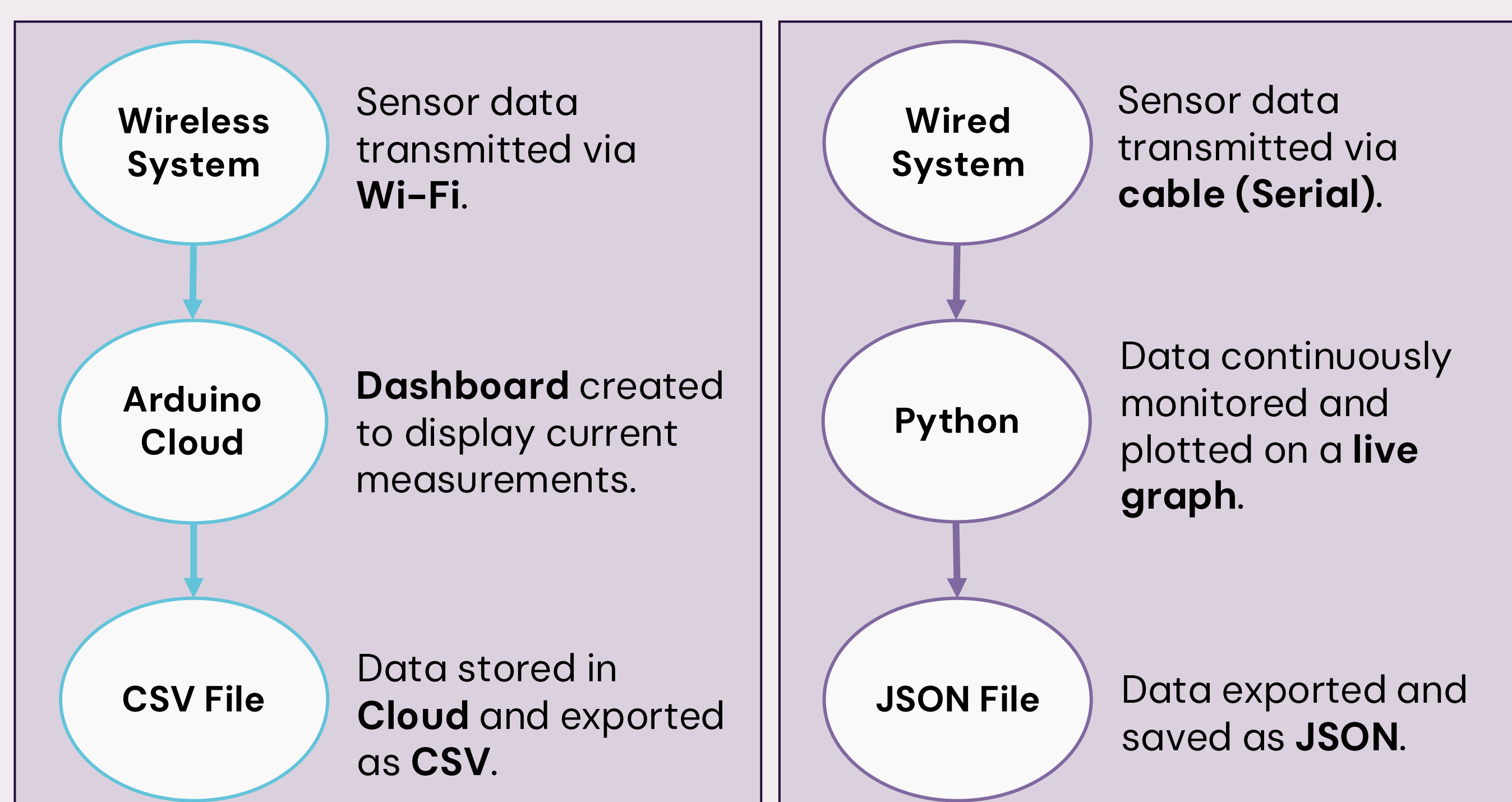


Figure 2: Method of data transfer by wired and wireless systems. The wireless system uses Arduino Cloud for data storage, and data collected throughout the day is exported as a CSV file for each sensor and variable. The wired system uses Python as a medium and outputs the data as an array which is then extracted as a JSON file for further analysis.

REFERENCES

[1] M. Marigo *et al.*, "Thermal comfort and productivity in a workplace: An alternative approach evaluating productivity management inside a test room using textual analysis," *Building and Environment*, vol. 245, p. 110836, Nov. 2023, doi: <https://doi.org/10.1016/j.buildenv.2023.110836>.

[2] ASHRAE. Thermal sensation scale, 2021.

Methodology

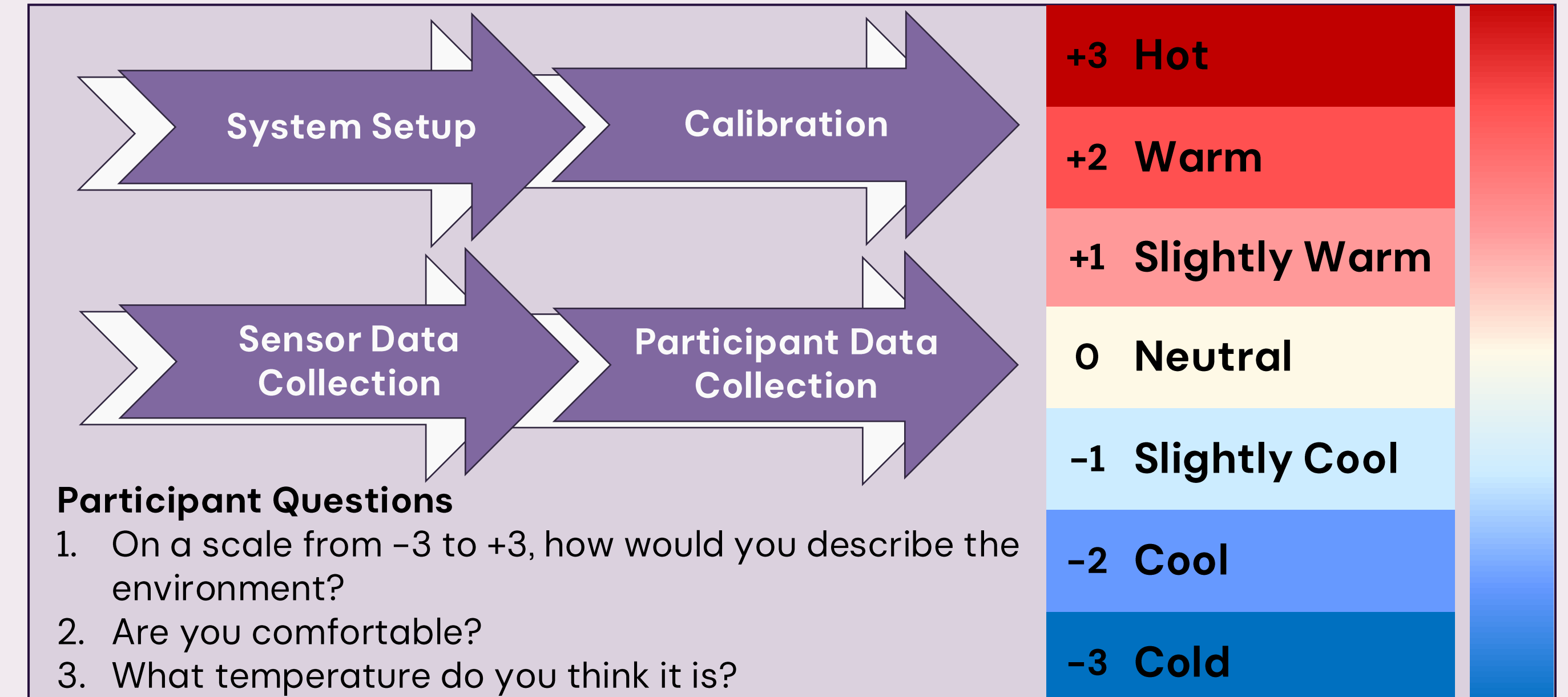


Figure 3: Experimental method pipeline. The questions use the PMV model [2] to compare an individual's subjective sensation to the environment.

Experimental Results

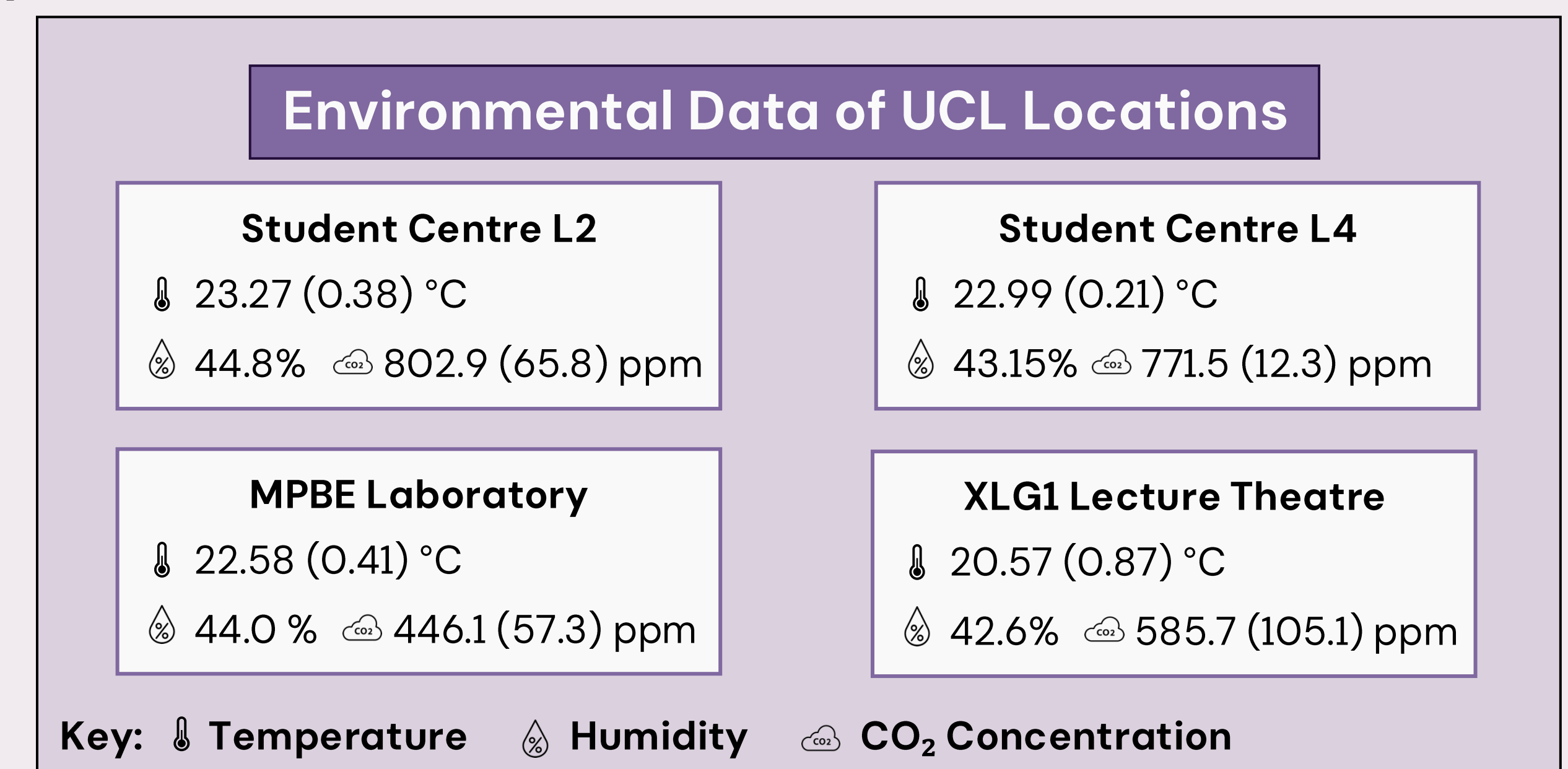


Figure 4: Summary of the average temperature and humidity measurements recorded by the wireless system in conjunction with the average CO₂ measurements recorded by the wired system.

Discussion

- Peak CO₂ in High-Traffic Zones**
 - The highest CO₂ levels were recorded in **Student Centre L2**, peaking at 1047 ppm.
 - This is due to factors such as **high occupancy, continuous foot traffic and limited ventilation**.
- Temperature Variance & Congestion**
 - The highest temperature of 23.27 °C was recorded in the **Student Centre L2**, due to **congestion during midday**.
 - Lecture Theatre** showed the greatest temperature **variation** due to **early morning** data collection.
- Overall Environmental Stability**
 - Humidity levels were **lowest in spacious locations** and higher in areas with poor ventilation.
 - Overall, the measurements suggest the **Laboratory** to be the **most thermally-stable** environment, but this needs further validation.

Subjective Survey Data

- Student Centre:** 71.4% of participants reported feeling 'slightly warm' (+1 on the PMV scale).
- Lecture Theatre:** Responses were highly varied, with 47.6% feeling 'neutral' (0), 28.6% 'slightly warm' (+1), and 23.8% 'slightly cool' (-1).
- Discomfort Levels:** 14.3% of participants reported feeling uncomfortable in the Student Centre, compared to 23.8% in the Lecture Theatre.

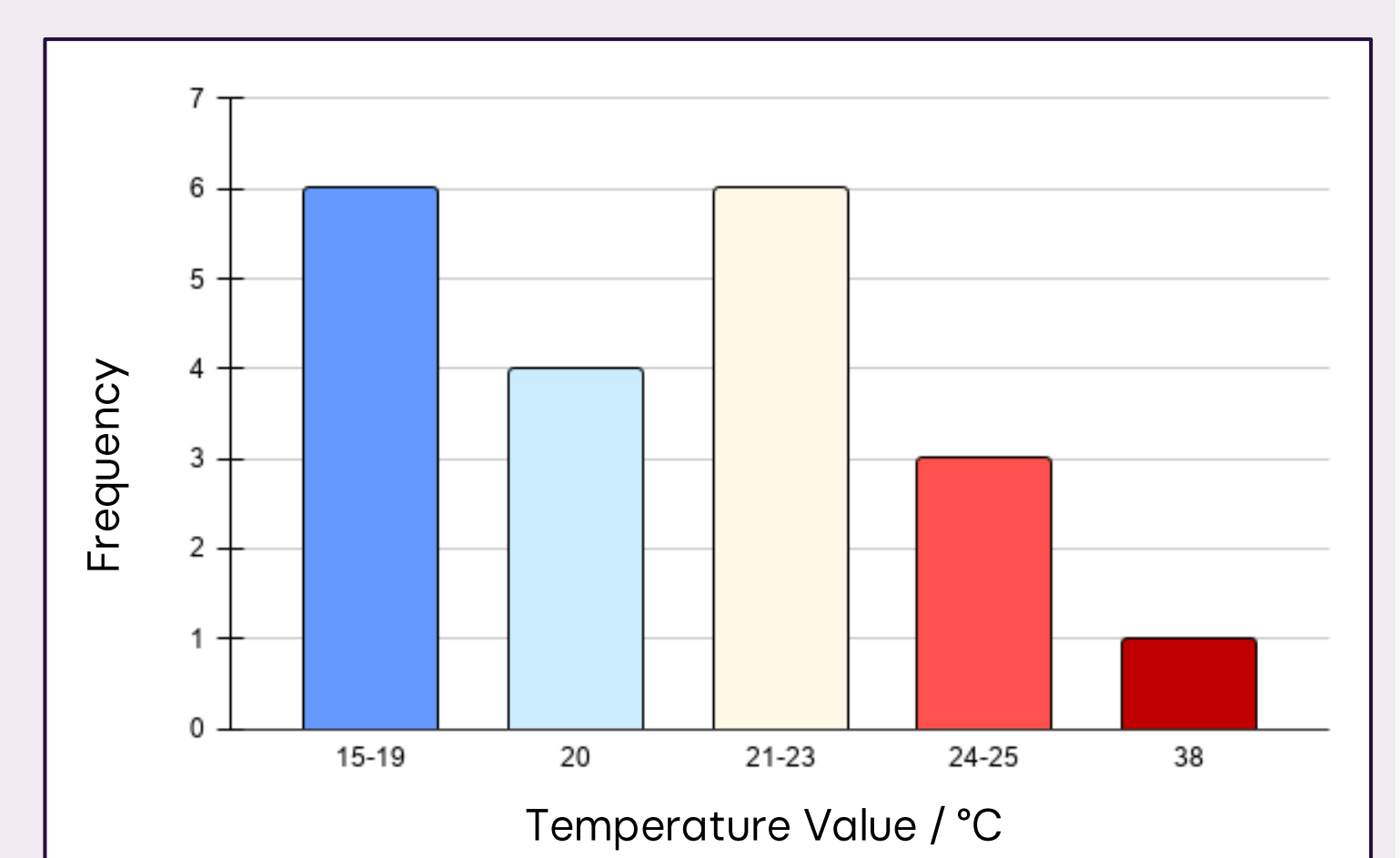


Figure 5: Frequency distribution of subjective ambient temperature estimates reported by occupants in the XLG1 Lecture Theatre. Column colours visually represent the temperature spectrum, from cooler (blue) to warmer (red).

Main Conclusions

- The variation of student temperature estimates highlights the **discrepancy between perceived temperature and measured values**, particularly in the Lecture Theatre.
- Despite higher CO₂ levels in the Student Centre and lower average temperatures in the lecture theatre, subjective responses indicate **greater discomfort in the lecture theatre** suggesting that **discomfort cannot be attributed to a single environmental variable**.
- Architectural features such as **windows and ventilation** created measurable differences between L2 and L4 of the Student Centre.
- The laboratory emerged as **the most thermally stable environment** but this requires further validation at periods with higher occupancy.

Evaluation

Our project successfully produced 2 systems that can measure environmental data across multiple locations. However, the subjective data used to correlate this to thermal sensation was severely limited by a small participant sample and one dataset per location.

Future studies should incorporate a greater number of locations to validate the data on temperature perception. Also, integrating the wireless system into a wristband structure could allow the validation of skin temperature data against human perception.